

Error localization kya hota hai ?

Error localization = galti *kahan* hui ye pehchanna

Bug is in line 27, not in main()

Error localization = line 27

LLMs normally sirf itna bolte hain:

“Program wrong hai”

Paper bolta hai:

“LLMs ko line number dena sikhana padega.”

Agar tum **first wrong step** nahi pakadte:

- Tum baad ke steps fix karoge
- But root galti wahi rahegi
- Result: same wrong answer, confidently

Most self-correction fails because models localize error too late.

Reasoning ko steps me todna (MOST IMPORTANT)

Step 1:

Step 2:

Step 3:

Step 4:

Answer:

Agar steps hi nahi honge, to localization impossible hai.

“Kaunsa step galat lag raha hai?”

“Pehla step kaunsa hai jo galat hai?”

Baad ke steps usi pe depend karte hain

- Agar first galti bach gayi → sab bekaar
isi ko bolte hain **clean prefix**.
- Verification ≠ Localization (yeh confusion hota hai)

Bahut log confuse karte hain:

Verification: answer sahi hai ya nahi?

Localization: galti kis step me hai?

Answer wrong ✗ → verification
Step 3 wrong ✗ → localization
LLMs verification thoda-bahut kar lete hain

Localization bina structure ke nahi hota
Once error mil gaya:

Galat approach:
“Sab kuch dobara likho”
Sahi approach:

“Is step se pehle tak sab sahi tha”

Then:

- Us step ke baad ka sab hatao
- Wahi se naya path try karo

Yehi Thought-ICS ka core loop hai.

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Here is my solution broken into steps.  
The final answer is incorrect.  
Identify the FIRST step where reasoning goes wrong.  
Explain why.
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Error might not be in model:

- Dataset bias
Label noise
Feature leakage
Evaluation metric
- Data → Cleaning → Features → Model → Eval
- IDEA EARLIER
- “LLMs self-correct nahi kar sakte”
- NOW: “LLMs correct tab nahi kar pate jab unhe pata hi nahi hota galti kahan hui.”

Galti sudharne se pehle, galti ki jagah fix karo.
CORE IDEA HAI

